The LAN-Only Policy

Which requests a node accepts from the local network only, how the tools decide that a request is local, which features enforce the rule, and which features only label a request as local without blocking it.

LOCAL NETWORK · TAILSCALE · ENFORCED · LABELS ONLY

SUMMARY · READ-ONLY REVIEW

The rule is set feature by feature

The tools do not share one LAN gate. Each feature decides for itself whether a request must be local, and each uses its own test. The servers listen on every interface, so the rule is applied in code, not by a firewall that the tools configure.

- **Listening.** The Dashboard and wifimon bind to `0.0.0.0`, so they accept connections on every interface, including the WAN side when the node has one.
- **Gating.** A feature that must be local checks the request's source address before it does its work.
- **Labelling.** Some views only report whether a client looks local. They do not refuse anything.

Read the book's wording with that split in mind. "LAN only" is true of a named feature, not of a whole tool.

The address test

2.1 The standard test

Most tools parse the peer address with Python's `ipaddress` module and accept it when it is one of:

- `is_loopback`: the node itself (`127.0.0.0/8`, `::1`)
- `is_private`: the RFC 1918 ranges and the IPv6 unique-local range
- `is_link_local`: the self-assigned `169.254.0.0/16` and `fe80::/10` ranges

IPv4-mapped IPv6 addresses (`::ffff:a.b.c.d`) are unwrapped first, so a mapped LAN address is treated like its IPv4 form.

2.2 Tailscale is added in two places

Node Talker (`_TAILNET`) and the SysMon Node Listen view (`_NL_TAILNET`) also accept `100.64.0.0/10`. That range is the carrier-grade NAT block that Tailscale uses, so a device on the node's tailnet counts as local there. No other tool adds it.

The address test is not the same as a logged-in user. A tailnet device that reaches a gated feature still needs the feature's own checks, such as a login, where those apply.

2.3 What a proxy does to the test

A request that arrives through a proxy or a tunnel comes from the proxy's address, which is often local. The tools therefore check for proxy headers as well (Part 4 of this series, Security hardening). A request that has a proxy header is refused, even when its address passes the test.

Where a request is refused

3.1 wifimon control actions

The control gate (`_control_gate`) runs before any change. It checks, in order:

1. The session is logged in (401 otherwise).
2. The source is loopback or private, and is not the unspecified address. The request has no proxy header (403, "Changes only work from this machine or the local network, not through a proxy.").
3. Changes are switched on, by the install-enable file, owned by root (403 otherwise).

The gate covers only the control paths: radio, device disconnect, switch, mode, forget, service restart and install start. Other paths are not gated by the LAN test.

3.2 44helper terminal

The terminal check (`_term_client_ok`) refuses any request with a proxy or Cloudflare header. It accepts only loopback, link-local and private addresses, and refuses the rest with the message "The terminal only works from your LAN." Enabling the terminal (`_term_can_enable`) also needs root and the terminal-enable switch.

3.3 Node Talker outside access

Node Talker treats a peer as outside (WAN) when the request is proxied or its address is not local. Outside requests are refused on the gated routes with 403, "Node Talker is not open to the internet (allow_wan is off)," unless the `allow_wan` setting is on. The default is off, so Node Talker is not open to the internet until an operator turns it on.

3.4 The Dashboard local API

The Dashboard's local API accepts only loopback peers, and it also needs the local key. This is the strictest rule in the set: the local API is not reachable from the LAN at all (see the security summary, Part 3).

Where "local" is only reported

FEATURE	WHAT IT DOES WITH THE TEST	BLOCKS ANYTHING?
SysMon Node Listen client list	Marks each connected client with a `lan` flag and a `local` flag for the loopback addresses	No
pibrary `access_info`	Describes how the viewer reached the page: this Pi, Tailscale or VPN, a LAN address, or unknown	No
instmon router link	Offers the router's web address only when the client is on the gateway's LAN network	No. It decides whether a link is shown
Dashboard phone check	Reports whether the phone's Pi address is a private address	No

These are useful to the operator, but they are not access rules. The book must not describe a label as a control.

What each tool enforces

TOOL	LOCAL TEST	ENFORCED WHERE
Dashboard (pi02w)	Loopback only for the local API; LAN labels elsewhere	Local API only
SysMon (pi02w)	Standard test, plus Tailscale in Node Listen	Labels only in the code read
wifimon	Loopback or private, with the proxy header refusal	Control paths (`_CONTROL_PATHS`)
44helper	Loopback, link-local or private, with the proxy refusal	Terminal and its enabling
Node Talker	Standard test, plus Tailscale	Gated routes, unless `allow_wan` is on
instmon	Gateway LAN network for the router link	Not a gate. Other routes need a session
pibrary	Standard test, for labels	Labels only in the code read

What the LAN-only policy means in practice

6.1 "Local" means the whole home network

A private address is any device on the LAN, not one trusted person. The policy keeps the internet out, but it does not separate household devices from each other. The book should say that plainly.

6.2 The tunnel is the main exception

A tunnel that brings internet traffic onto the LAN, such as a Cloudflare tunnel, is refused by the proxy headers. A tunnel that places the remote device on the LAN, such as Tailscale, is accepted by design. Operators who want the tunnel refused must rely on the feature's own login.

6.3 The rule is set in code

Changing who is local means changing the code of each tool. There is no setting that moves every feature to a stricter policy at once.

6.4 Not checked

- Every route in the Dashboard, SysMon and instmon for a local test. Only the gates named above were read.
- The full list of Node Talker's gated routes.
- How the wifimon install-enable file is set on a running node.
- Whether the firewall rules the tools open are restricted by source address.